



AR and VR for military training needs (Credit: www.hill.af.mil)

revolutionise everyday lives. AR harnesses the ability of ML to learn and remember imagery. Similarly, AR apps created for mobiles utilise AI and ML capabilities of smartphones.

Immersive head-mounted displays (HMDs) are primarily used to experience AR and VR. AR HMDs drive enterprise usage as hands-free tools for business process improvement and training.

Realty industry

Housing.com has launched a first-of-its-kind innovative VR-based solution called RealtyVision 2.0 for 3D presentation of real-estate projects. The portable product offers 3D walkthrough support, and can be controlled from both the buyer's as well as the seller's end. It reflects the consumer's real-time journey via touchscreen. This feature enables the seller to guide, explain and navigate during the digital tour through convenient remote/pointer-based interactions.

The state-of-the-art technological solution offering integrated 3D experience works without an Internet connection, easily facilitating customer visits. It allows the builder to sync all data through the cloud, and update fresh information and new projects as per need.

Ravi Bhushan, group CPTO, Housing.com, PropTiger.com and Makaan.com, says, "VR is moving from being a novelty to a necessity in the real-estate sector. Homebuyers, developers and real-estate agents are demanding a better and more efficient way of discovering and showcasing properties. We believe that this innovation will empower consumers to make smarter purchasing decisions more efficiently."

Manufacturing industry

3D AR and VR platforms have applications in multiple domains for developing real-time simulations that can create virtual environments. Such platforms are transforming the way OEMs execute designing, marketing, training, autonomous driving and human-machine interactions. These enable OEMs to innovate, deploy and test human-machine interfaces (HMIs) for the next generation of vehicles through a real-time 3D-simulated environment.

OEMs are witnessing the ability to conceptualise autonomy to advanced levels of safety and smartness without undergoing physical testing to keep their business costs down. Customers can also configure or experience in VR before physical cars are available in retail.

Security services

AR and VR can solve complex problems of the digitising world. With constant innovation and integration of AR/VR, Indian defence systems can enhance their strategy and use these technologies for such visual information as horizon line, aircraft speed and altitude. Anumukonda Ramesh, country head - India, Unity Technologies, says, "Defence and PSU sectors are the biggest users of simulation and visualisation for training, learning and product designing."

AR military systems can test any weaponry before use; for example, ARC4 connects to military satellites and drones as well as highlights dangers and threats in AR mode.

VR, AR and MR require new computing platform designs to replicate the real environment. These use advanced processing, sensing and display technology, and are ideal for defence sector applications to gain experience of dangerous environments from the convenience of a training room. Such applications can put trainees in different situations and environments for teaching awareness and upskilling to provide valuable experiences.

Retail industry

AR and VR have the potential to revolutionise customer experience and drive revenues by individualising retailer offers and enabling customers to visualise products in different settings. Retailers can also enhance the understanding of information through advanced graphical visualisation and simulation technologies.

Pilots and implementation examples include VR tours by Tesco, Alibaba's full VR shopping experience, Adidas' VR videos to promote its outdoor clothing collection and FR Tech-iLUK for personalised styling. With AR-enabled platforms, buyers can also interact with the product(s) they want to buy to better understand them and explore their functionalities.